

Nobli

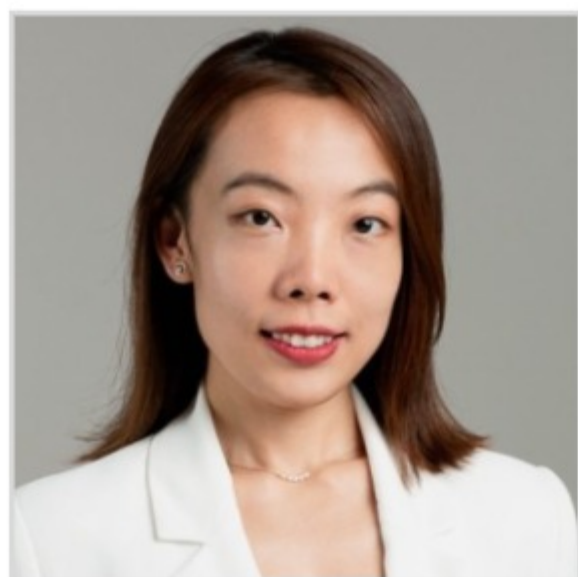
**Research at Nobel speed.
Growth at your pace**



Nobli

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Problem

THE SCALE OF ATTRITION

726,000

Students Facing 30-50% Attrition



Every year, nearly half of all doctoral candidates in the EU risk leaving their programs.



THE ROOT CAUSES

MAIN REASONS FOR DROPPING OUT



Poor Work-Life Balance

High institutional pressure and excessive workloads frequently lead to chronic stress and burnout.



Broken Supervisor Relationships

Power imbalances and a lack of transparent mentorship create significant barriers to degree completion.



Metric-Driven Publication Pressure

Evaluation systems prioritize quantity over research quality, isolating students in high-pressure environments.

Life change with Nobli



Value 1

Automate the whole
research flow



Value 2

Amplify the mind.
Grow into the
researcher you're
meant to be.



Value 3

It's NOT a paper
maschine, but a
mentor mashine.
Nobel-level Professor
for 24 h.

Feature Scope & Success

	End-to-End Paper Generation	Agent Self-Growth	Recursive Improvement	Human-Centric Growth
Nobli	✓	✓	✓	✓
AutoResearch (Karpathy)	✗	✗	✓	✗
AutoResearch Claw (UNC)	✓	✓	✓	✗
AI-Researcher (HKU)	✗	✗	✗	✗
Feynman (Companion)	✗	✗	✗	✓
AutoResearch-DS (DeepSeek)	✗	✗	✓	✗
Lemma (Analemma)	✗	✗	✗	✗

Methodology & Architecture

Fig 1: System Overview

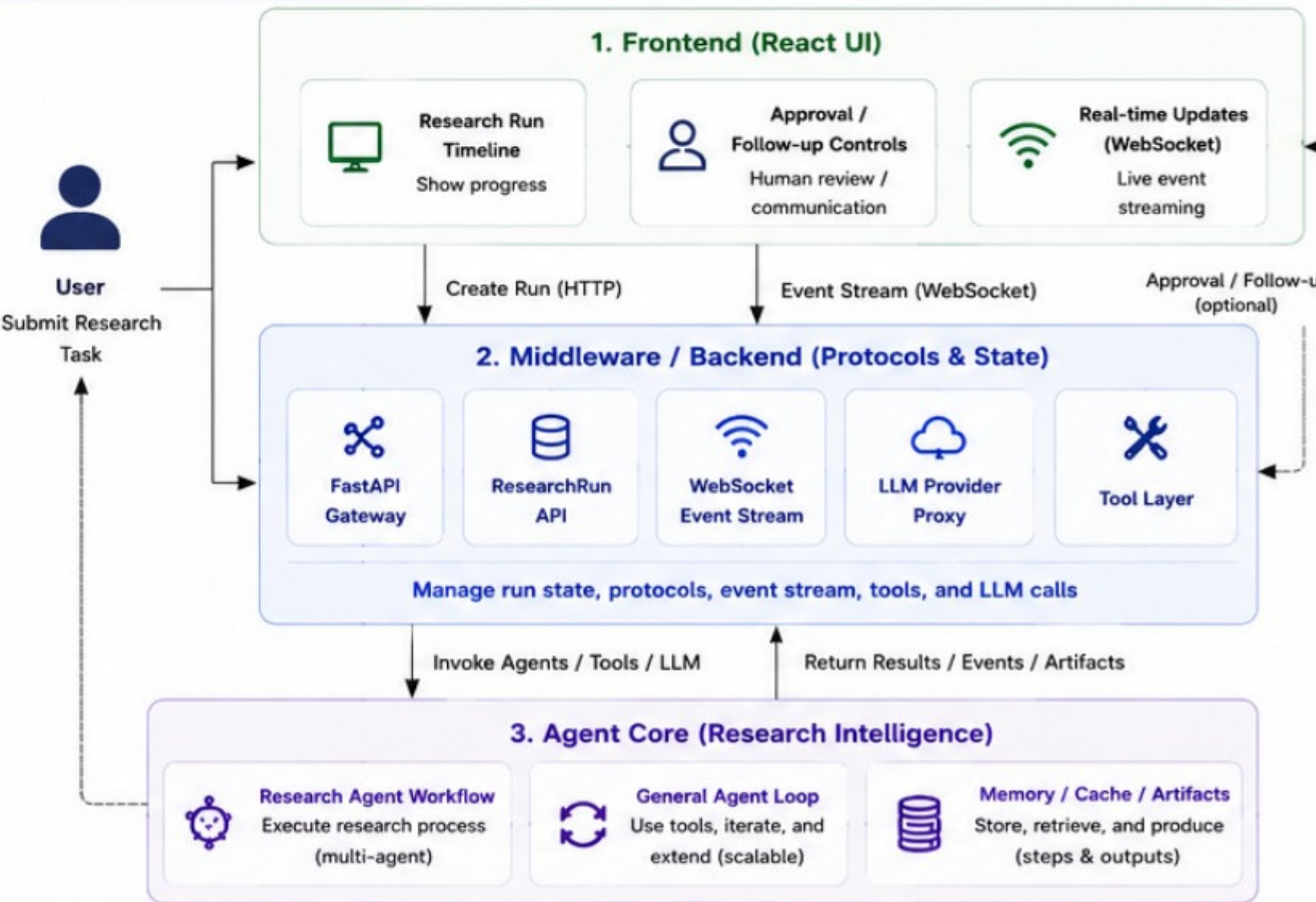


Fig 2: Core Agent Catalog (Roles & Outputs)

Agent	Primary Responsibilities	Core Outputs
Prepare Agent (Planning Agent)	- Search related papers & code - Build research background	- Related paper list - Reference code / datasets
Survey Agent (Research Agent)	- Read & understand papers - Extract key concepts, methods, metrics	- Concept map / knowledge structure - Data table & metric list
Coding Plan Agent (Experiment Design Agent)	- Design experimental plan - Generate executable plan	- Experiment plan (structured) - Evaluation metrics & baselines
ML Agent (Implementation Agent)	- Implement models & code - Make code runnable	- Source code / project - Training scripts / configs
Judge Agent (Evaluation Agent)	- Evaluate results - Provide feedback & suggestions	- Evaluation report / feedback - Improvement suggestions
Paper Generation Agent (Writing Agent)	- Write paper content - Organize experiments & results	- Paper (main text) - Figures / tables / appendix
Memory Update Agent (Memory Agent)	- Record key findings & learnings - Update long-term memory	- Research memory entries - Experience summary

Fig 3: Research Run Workflow (with Human Gates)

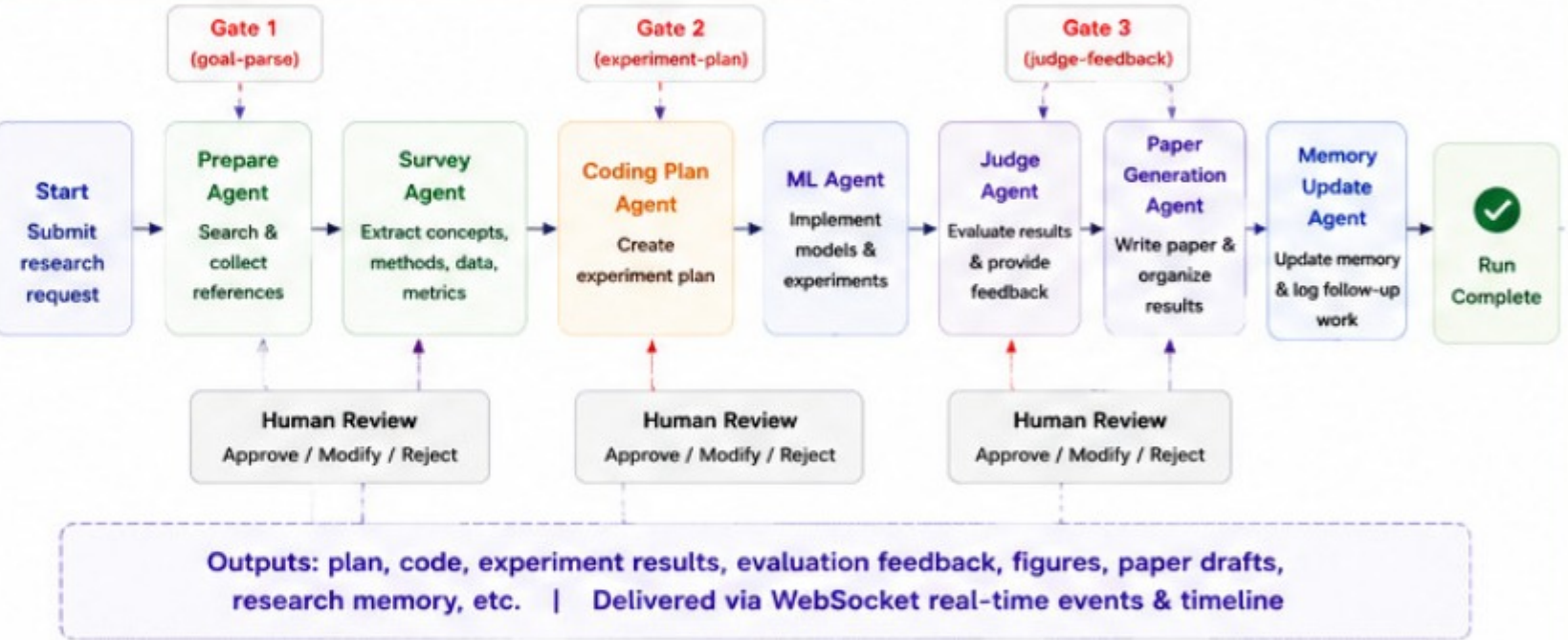


Fig 4: Self-Grow & Recursive Improvement (Dual Loops)

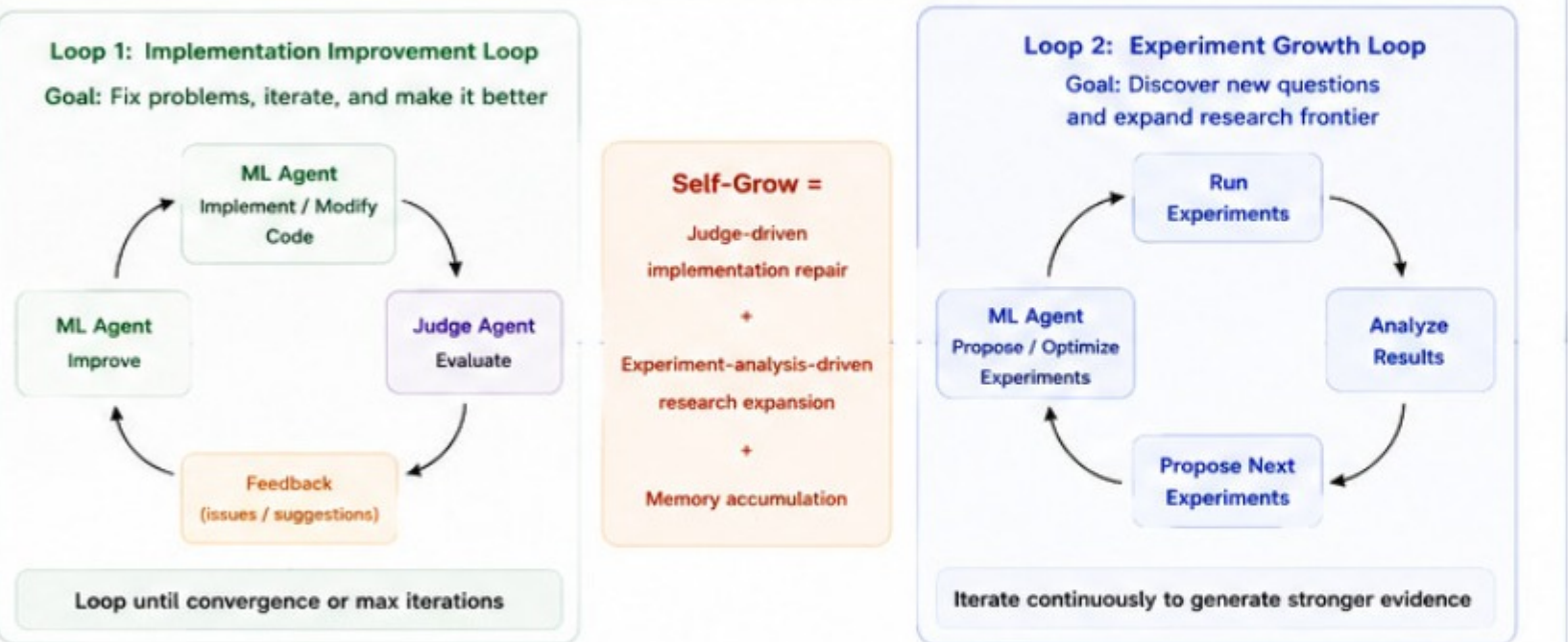
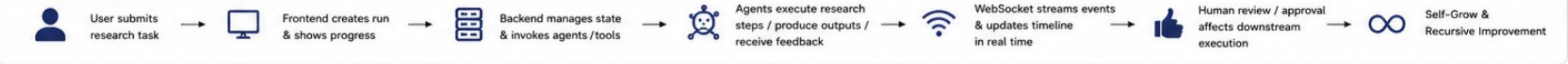


Fig 5: End-to-End Summary (Data Flow)



Demo

Show time!

Result & Matrics

Case Prompt:

I want to build a better CoT algorithm but not sure what is the idea

- **End-to-end research output**
Built **1 complete agentic research workflow**
- **Research contribution**
Generated **1 submission-ready paper draft**, including abstract, introduction, experiment plan, citation support, and reviewer-style feedback.
- **Time saving**
Reduced early-stage paper development from **70 hours to 3 hours:**
~96% time saved.
- **Self-growing**
 - **System self-growth**
Orchestrated **5 specialized agents** as continuous learning signals.
 - **Human-in-the-loop**
Added **3 approval gates**: experiment revision, reviewer feedback, final paper draft.
 - **Self-growing researcher**
Provided **reviewer feedback + frontier paper recommendations + conference alerts**, helping users learn expert research thinking while working.

When Does Chain-of-Thought Hurt? Budget-Aware Prompting for Reasoning Under Tight Output Limits

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Abstract

Chain-of-thought (CoT) prompting is widely assumed to improve reasoning, but it spends output tokens before reaching an answer. We study reasoning accuracy under a fixed output-token budget and find a robust non-monotonic effect: when the budget is tight, vanilla CoT performs *worse* than answering directly, because the model is truncated mid-reasoning and never emits a final answer. We characterize this “CoT collapse” regime and propose *budget-aware CoT*, a simple answer-first prompting strategy that commits to an answer on the first line and then adds compressed symbolic justification, so a truncated reply still contains a usable answer. On a suite of multi-step reasoning problems evaluated with `gpt-4o-mini`, budget-aware CoT dominates both direct answering and vanilla CoT across the entire budget range, with the largest gains (up to +30 accuracy points) in the tight-budget regime where vanilla CoT collapses. The optimal prompt was discovered by an autonomous keep-or-revert experiment loop, illustrating a path toward self-directed prompt research.

1 Introduction

Large language models reason more accurately when prompted to “think step by step” [1]. This benefit, however, is usually measured with generous decoding budgets. In practice, output length is frequently capped: latency-sensitive products, batched serving, and cost controls all impose tight `max_tokens` limits. Under such caps the cost structure of CoT changes qualitatively—tokens spent narrating reasoning are tokens unavailable for the answer itself.

We ask a simple question: *how does the best prompting strategy change as a function of the output-token budget?* We evaluate three strategies—direct answering, vanilla CoT, and a budget-aware variant we propose—across a sweep of budgets on multi-step reasoning problems. We find that vanilla CoT is non-monotonically related to the budget relative to direct answering: it is *worse* at small budgets and better at large ones, crossing over at an intermediate budget. We then show that a small change to the prompt—emitting the answer first—removes the collapse and yields the best accuracy at every budget.

Contributions. (i) We document a CoT-collapse regime under tight output budgets and locate the crossover point against direct answering (Section 4). (ii) We propose *budget-aware CoT*, an answer-first prompt that is robust to truncation (Section 3). (iii) We report that the strategy was found by an autonomous experiment loop that keeps or reverts prompt edits based on measured accuracy, suggesting a practical recipe for self-directed prompt optimization.

39th Conference on Neural Information Processing Systems (NeurIPS 2025).

Final paper

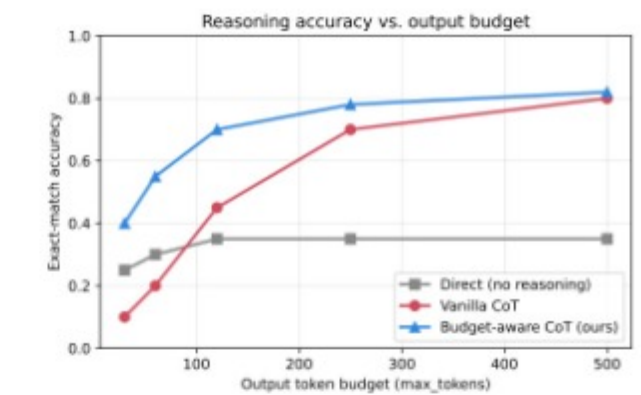


Figure 1: Exact-match accuracy versus output-token budget. Vanilla CoT (red) is below direct answering (gray) at small budgets and crosses over at larger ones; budget-aware CoT (blue, ours) dominates throughout.

Limitations. We evaluate one model and a modest problem suite with exact-match scoring; the precise crossover budget will depend on model, task, and tokenizer. Answer-first prompting may forgo some accuracy that fully elaborated reasoning would attain at very large budgets, though we do not observe this in our range. Finally, the autonomous loop optimizes a single scalar and could overfit a small evaluation set; larger held-out suites are future work.

Conclusion. Under tight output budgets, prompt strategy should be chosen *as a function of the budget*. Budget-aware, answer-first CoT is a simple, robust default, and an autonomous keep-or-revert loop can discover it without human prompt engineering.

References

- [1] J. Wei et al. Chain-of-thought prompting elicits reasoning in large language models. In *NeurIPS*, 2022.
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- [3] X. Wang et al. Self-consistency improves chain of thought reasoning in language models. In *ICLR*, 2023.
- [4] S. Nayab et al. Concise thoughts: Impact of output length on LLM reasoning and cost. *arXiv preprint*, 2024.
- [5] Y. Zhou et al. Large language models are human-level prompt engineers. In *ICLR*, 2023.

Thank you!